GPS Driver and Data Analysis Report

Introduction / Objective

In this lab, the aim was to develop a driver for a GNSS puck to handle $GPGGA formatted messages, convert geographic coordinates to UTM, and share this data via a custom ROS message. After developing the driver, we test it by collecting and analyzing GPS data in stationary and moving scenarios.

Steps:

A driver was created that reads data from the GNSS puck via a serial connection. This allows the driver to continuously receive and process incoming data from the puck. The driver focuses on parsing $GPGGA messages, which contain information such as latitude, longitude, and altitude. By splitting these messages, we can extract the needed data. Once we have the latitude and longitude, we convert these to UTM coordinates using the utm library. This conversion gives precise easting and northing values, along with the zone information. To communicate the parsed and converted data, we created a custom ROS message (GPSmsg.msg). The driver publishes this custom message to the '/gps' topic, ensuring that other components of our system can access the data. The header includes the timestamp from the GPGGA message to maintain synchronization with the GNSS data, and the frame ID is set to "GPS1_Frame." We made the driver script (driver.py) flexible by allowing it to accept the serial port path as an argument. This way, it can easily be adapted to different setups.

Stationary Data

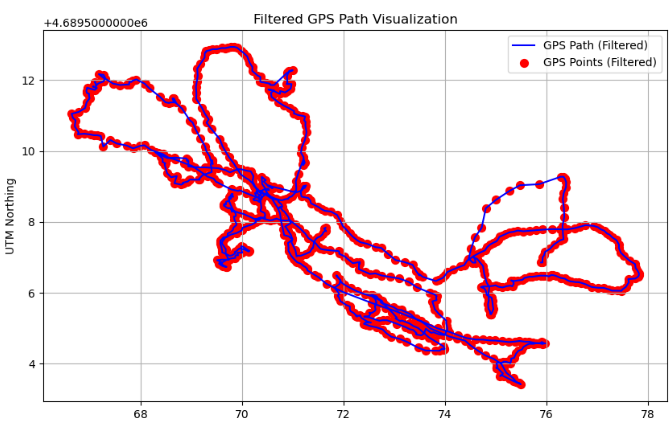
This data was collected while remaining stationary outdoors (outisde Snell Library) for 10 mins. This data was saved in a rosbag file named stationary_data_0.bag.

Walking Data

This data was collected while walking, one normally and another walking a bit faster outdoors (from Ruggles to the 2nd Bridge) and was saved in a rosbag file named walking_data_1.bag and walking_data_fast_0.bag respectively.

Data Analysis

1. Stationary Data Analysis



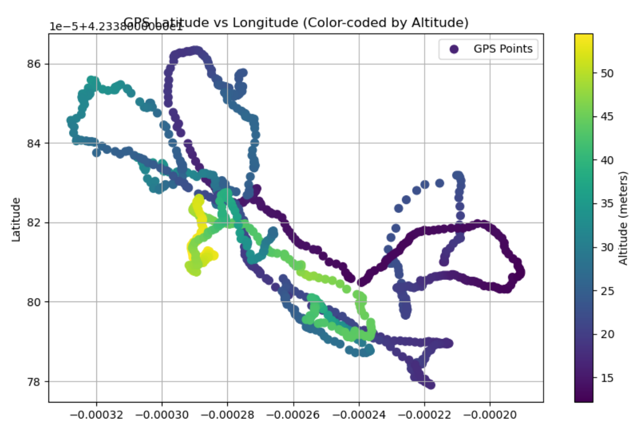
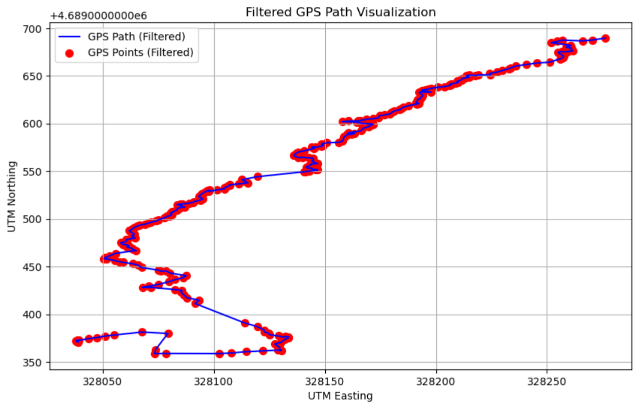
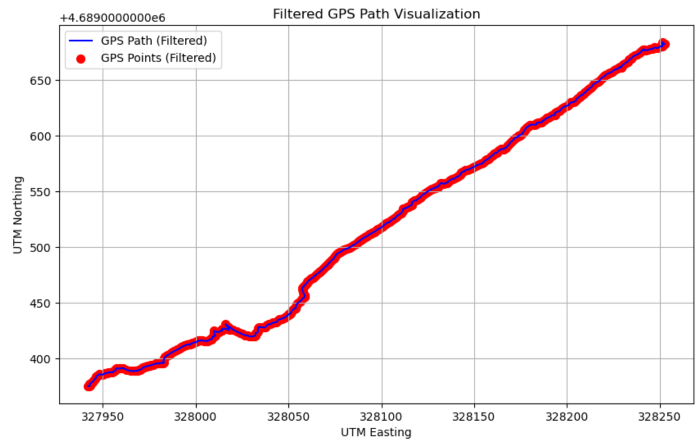
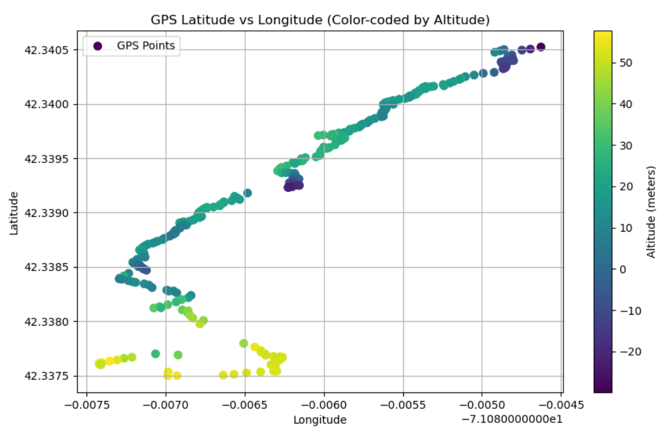


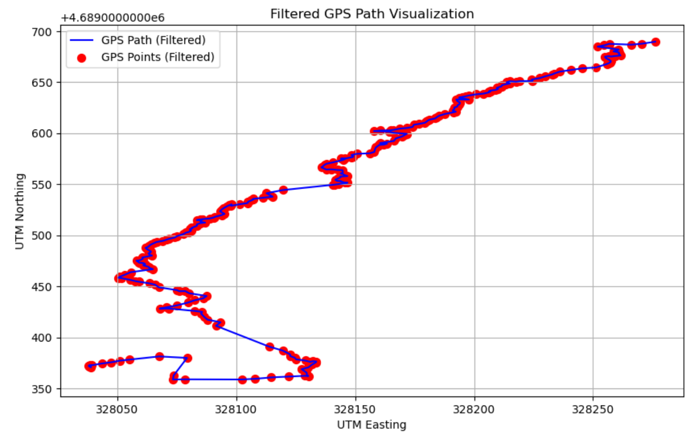
Image on left shows the Northing V/s Easting and the rigth one is of Latitude vs Longitude and Altitude on the color scale. The stationary data is plotted as above. We can see that even when we remain at a stationary position the coordinates don't stay at a point but wonder over. The reasons likely to cause these could include signal reflections from nearby buildings, atmospheric conditions, and satellite positionings

2. Walking Data Analysis









The above two graphs are plotted when walking normally along a straight line and the below two are when walking fast. We see a significant difference between the two as while walking slowly has better accuracy and precision as compared to the one when walking faster. The major factor for this, I think, could be that when walking faster, GPS gets less time at a spot and thus less data to pinpoint that location. But when we walk slower, it records more data from one location but also keeps changing the position thus reducing err

The reason the walking data appears more accurate than the stationary data lies in the way GPS signals work. When you move around, the GPS receiver on your device gets a chance to gather signals from different positions, which helps to balance out various sources of error. As you walk, your changing location allows the GPS to average out inaccuracies that might come from things like atmospheric conditions, the positions of the satellites, or reflections of the signals off buildings and other structures. This constant movement helps the device get a clearer, more accurate picture of your location. On the other hand, when you're standing still, the GPS receiver stays in the same spot, and any errors in the signal are more pronounced and persistent. Moving around effectively helps to "smooth out" these errors, leading to more reliable and accurate GPS data.